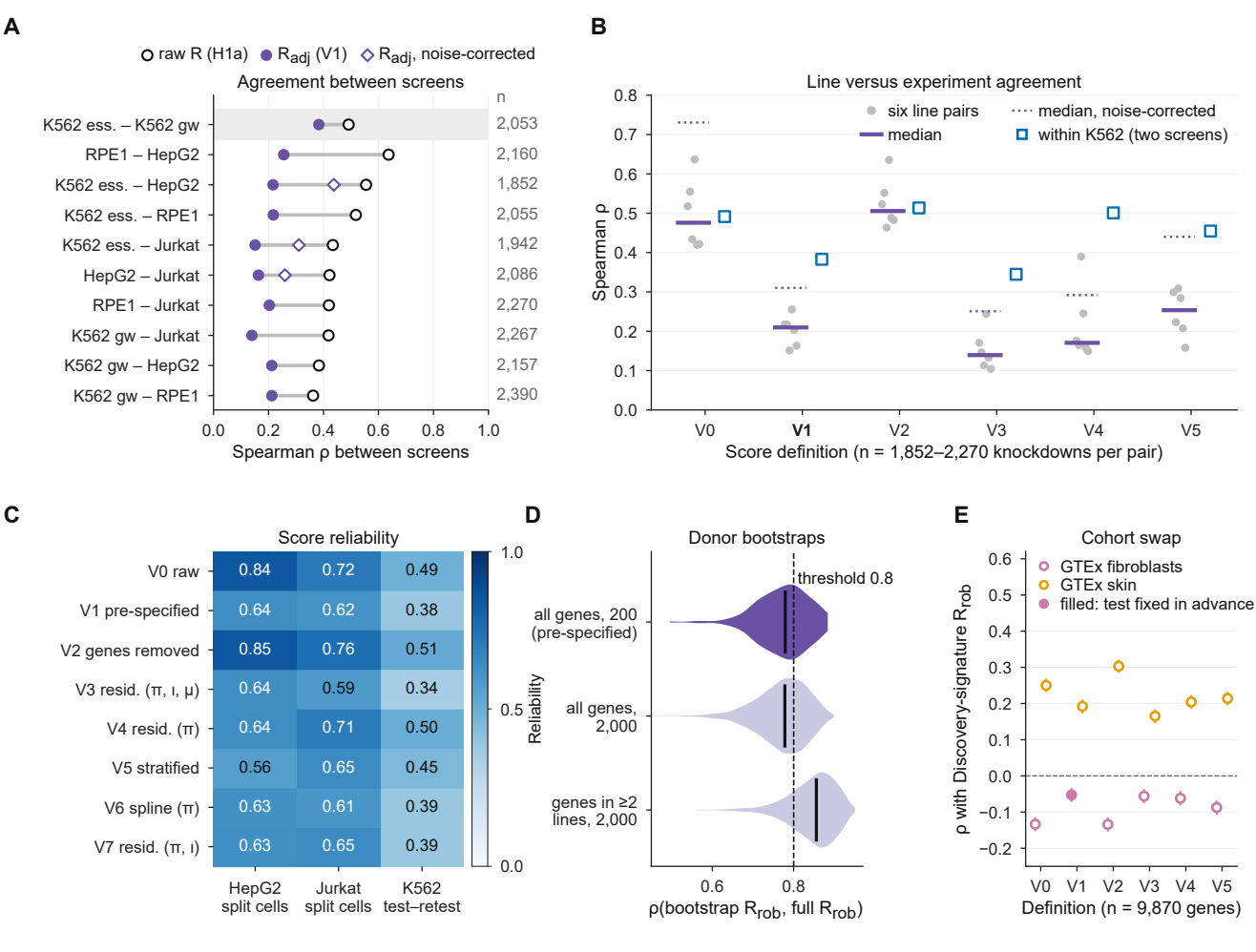

A
raw R (H1a)
R_{adj} (V1)
R_{adj}, noise-corrected
Agreement between screens
n
K562 ess. – K562 gw
2,053
RPE1 – HepG2
2,160
K562 ess. – HepG2
1,852
K562 ess. – RPE1
2,055
K562 ess. – Jurkat
1,942
HepG2 – Jurkat
2,086
RPE1 – Jurkat
2,270
K562 gw – Jurkat
2,267
K562 gw – HepG2
2,157
K562 gw – RPE1
2,390
0.0
0.2
0.4
0.6
0.8
1.0
Spearman ρ between screens
B
Line versus experiment agreement
six line pairs
median
median, noise-corrected
within K562 (two screens)
Spearman ρ
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
V0
V1
V2
V3
V4
V5
Score definition (n = 1,852–2,270 knockdowns per pair)
C
Score reliability
V0 raw
0.84
0.72
0.49
V1 pre-specified
0.64
0.62
0.38
V2 genes removed
0.85
0.76
0.51
V3 resid. (π, ι, μ)
0.64
0.59
0.34
V4 resid. (π)
0.64
0.71
0.50
V5 stratified
0.56
0.65
0.45
V6 spline (π)
0.63
0.61
0.39
V7 resid. (π, ι)
0.63
0.65
0.39
1.0
0.5
0.0
HepG2 split cells
Jurkat split cells
K562 test–retest
D
Donor bootstraps
threshold 0.8
all genes, 200 (pre-specified)
all genes, 2,000
genes in ≥2 lines, 2,000
Reliability
0.6
0.8
ρ(bootstrap R_{rob}, full R_{rob})
E
Cohort swap
GTEx fibroblasts
GTEx skin
filled: test fixed in advance
ρ with Discovery-signature R_{rob}
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
V0
V1
V2
V3
V4
V5
Definition (n = 9,870 genes)